MULTILEVEL INHERITANCE IN JAVA

Experiment: Multilevel Inheritance

1. Objective

To understand and implement multilevel inheritance in Java, where a class inherits from another derived class, forming a chain of inheritance.

2. Task Description

Create a Java program using three levels of classes: College, Department, and Student. College contains college details, Department inherits College and contains department details, and Student inherits Department and contains student details. Create a Student object and access all three methods.

3. Step-Wise Execution

Step 1: Create the parent class **College** and define the method **collegeDetails()**.

Step 2: Create **Department** using **extends College**. It inherits the College method and defines **departmentDetails()**.

Step 3: Create **Student** using **extends Department**. It indirectly inherits College and directly inherits Department, and defines **studentDetails()**.

Step 4: Declare the public class **MultilevelInheritanceDemo** containing the main() method.

Step 5: Create an object **s** of Student using **Student s = new Student();**.

Step 6: Call **s.collegeDetails()**, **s.departmentDetails()**, and **s.studentDetails()**.

Step 7: The three messages are displayed in the output.

4. Algorithm

1. Start.

2. Define class College with collegeDetails().

3. Define class Department extending College with departmentDetails().

4. Define class Student extending Department with studentDetails().

5. Define main() inside MultilevelInheritanceDemo.

6. Create a Student object.

7. Call all three inherited/own methods.

8. Display the college, department, and student details.

9. Stop.

5. UML Diagram

College	collegeDetails()
↑	
Department	departmentDetails()

↑	
Student	studentDetails()

Inheritance relationship: College → Department → Student

6. Program Code

```
class College {
    void collegeDetails() {
        System.out.println("College: ABC Engineering College");
    }
}

class Department extends College {
    void departmentDetails() {
        System.out.println("Department: Computer Science");
    }
}

class Student extends Department {
    void studentDetails() {
        System.out.println("Student: Rahul");
    }
}

public class MultilevelInheritanceDemo {
    public static void main(String[] args) {

        Student s = new Student();

        s.collegeDetails();
        s.departmentDetails();
        s.studentDetails();
    }
}
```

7. Code Explanation

class College: This is the base/parent class. It contains college-related information.

collegeDetails(): This method prints the college name.

class Department extends College: Department inherits the members of College. Therefore, Department can use collegeDetails().

departmentDetails(): This method prints the department name.

class Student extends Department: Student inherits Department. Since Department already inherits College, Student can also access College's method. This creates three levels.

Student s = new Student();: This creates an object of Student. Through this object, methods from Student, Department, and College can be accessed.

s.collegeDetails();: Calls the method inherited from College.

s.departmentDetails();: Calls the method inherited from Department.

s.studentDetails();: Calls the method defined in Student.

Overall concept: The inheritance chain is College → Department → Student. This is called multilevel inheritance because inheritance occurs through multiple levels.

8. Output Screen

```
Output Screen
```

```
College: ABC Engineering College

Department: Computer Science

Student: Rahul
```

9. Result

The Java program was successfully implemented and demonstrates multilevel inheritance. A Student object accesses methods from Student, Department, and College through the inheritance chain.

10. Conclusion

Multilevel inheritance allows a derived class to inherit from another derived class. In this program, Student inherits Department, and Department inherits College. Thus, Student can access the methods of all three classes.